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Early Triassic palaeomagnetic results from the Ryeongnam Block, Korean Peninsula: the eastern extension of the North China Block

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SUMMARY

Greenish sandstones in the Early Triassic Nogam Formation of the Ryeongnam Block, Korean Peninsula were collected at 23 sites for palaeomagnetic study. A high-temperature magnetization component with unblocking temperatures of 670–690 °C was isolated from seven sites and yielded a positive fold test at the 95 per cent confidence level. The high-temperature component is interpreted to be of primary origin because the folding age is Middle Triassic. The Early Triassic palaeomagnetic direction for the Ryeongnam Block after tilt correction is $D = 347.1^\circ$, $I = 23.8^\circ$ ($\alpha_{95} = 5.5^\circ$). The palaeomagnetic pole (62.5°N, 336.8°E, $A_{95} = 4.7^\circ$) shows good agreement with the coeval pole for the North China Block, suggesting that the Ryeongnam Block has been part of the North China Block at least since Early Triassic times. A tectonic history of the Korean Peninsula includes obduction of the eastern part of the South China Block onto the central part of the Korean Peninsula in the Permian, with the Ryeongnam Block geographically isolated from the main part of the North China Block. Collision of the North and South China blocks commenced initially at the Korean Peninsula, and suturing of the two blocks progressed westwards.

Key words: Korea, palaeomagnetism, tectonics, Triassic.

1 INTRODUCTION

The Korean Peninsula is situated to the east of the major tectonic blocks comprising East Asia (North and South China blocks, see Fig. 1a). It is composed of three tectonic units with basement of Archean to middle Proterozoic age, namely the Nangrim–Pyeongnam, Gyeonggi and Ryeongnam blocks from north to south (Reedman & Um 1975; Cluzel *et al.* 1990) (Fig. 1b). The former two blocks are bounded by the Imjingang Belt, and the latter two by the Ogcheon Belt.

The Ryeongnam and Nangrim–Pyeongnam blocks have been correlated with the North China Block, whereas the Gyeonggi Block has been correlated with the South China Block (Cluzel *et al.* 1990). According to palaeontological data (Kobayashi 1967), the former two blocks have Cambro-Ordovician faunas with North China affinity, whilst the latter block has Cambro-Ordovician faunas with South China affinity.

Palaeomagnetic studies of the Ryeongnam Block in late Palaeozoic to early Mesozoic times have been carried out to confirm its affinity with the North China Block. Although several palaeomagnetic results have been reported from the Ryeongnam Block, their tectonic interpretations are not consistent. Doh & Piper (1994) interpreted the Ryeongnam Block as an eastern extension of the North China Block because of good agreement of Late Carboniferous to Early Triassic palaeomagnetic poles between the two blocks. Lee *et al.* (1997) gave Late Carboniferous to Early Triassic poles far removed from coeval poles for the North China Block; however, they claimed that the Ryeongnam Block is part of the North China Block because of the consistency of the inferred palaeolatitude of the Ryeongnam Block with the North China Block rather than the South China Block. Kim, Lee & Choi (1992) concluded that the Ryeongnam Block sutured into the South China Block in the Jurassic on the basis of Late Carboniferous to Early Triassic poles that lie near coeval poles from the South China Block.

Attempts to locate the source of the Ryeongnam Block from palaeomagnetic evidence have been hampered by severe remagnetization of late Palaeozoic–early Mesozoic sedimentary rocks within this block. Many authors observed persistent remagnetization of Cretaceous, Tertiary and Recent ages (Otofuji *et al.* 1986; Shibuya *et al.* 1988; Kim *et al.* 1992; Lee *et al.* 1997). Otofuji *et al.* (1989) and Doh *et al.* (1997) found only remagnetized palaeomagnetic directions. Because of this strong remagnetization, palaeomagnetic study in the Korean Peninsula has failed to accumulate sufficient reliable late Palaeozoic–early Mesozoic results to resolve the palaeogeographic affinity of this region.

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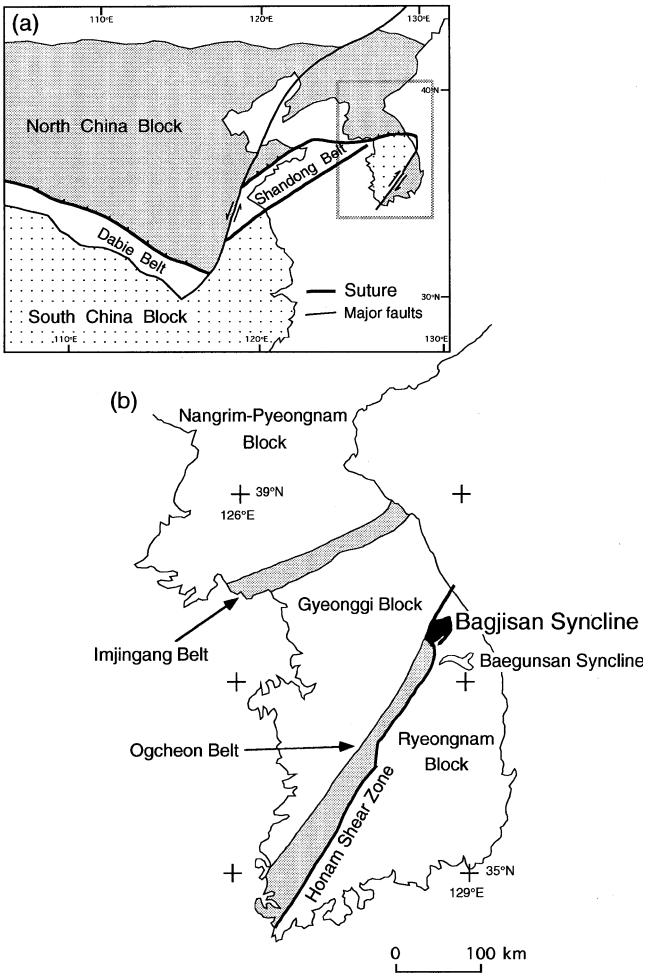


Figure 1. (a) Tectonic map of the Korean Peninsula and adjacent area. (b) Tectonic division in the Korean Peninsula. The Ryeongnam Block is concluded to be the eastern part of the North China Block in this study. The Bagjisan Syncline (study area) is situated in the northeastern part of the Ryeongnam Block.

This paper focuses on the tectonics of the Ryeongnam Block in Early Triassic times with a view to comparing the palaeomagnetic poles from the Ryeongnam Block with those from the North and South China blocks in order to test the affinity of the Ryeongnam Block. The target for this palaeomagnetic study is the Lower Triassic Nogam Formation in the Pyeongan Supergroup. This formation is of a strategic age because the Early Triassic palaeomagnetic poles from the North and South China blocks are far apart from each other (Gilder & Courtillot 1997). The study area is the Bagjisan Syncline located in the northeastern part of the Ryeongnam Block (Fig. 1b), an area where a fold test is available.

2 GEOLOGICAL SETTING AND SAMPLING

In the northeastern part of the Ryeongnam Block, the Late Carboniferous–Early Triassic Pyeongan Supergroup unconformably overlies the Early Cambrian–Middle Silurian Joseon Supergroup, and is unconformably overlain by the Late Triassic–Early Jurassic Daedong Supergroup.

The Pyeongan Supergroup is well exposed in the Bagjisan and Baegunsan synclines (Fig. 1b). The Pyeongan Supergroup is divided into four formations: the Hongjeom, Sadong, Gobangsan and Nogam formations in ascending order (Reedman & Um 1975; Lee 1987). The ages of the former three formations are determined by palaeontological data as Late Carboniferous, Early Permian and Late Permian, respectively (Kawasaki 1939; Cheong 1974). The Nogam Formation is assigned an Early Triassic age on the basis of stratigraphic and structural data (Reedman & Um 1975). The Hongjeom and lower Sadong Formations were deposited in shallow marine basins, while the upper Sadong, Gobangsan and Nogam formations were formed in non-marine conditions (Reedman & Um 1975). The Nogam Formation consists of green, red and grey sandstones, grey shales, and conglomerates, with a total thickness of 3000 m. The upper part of the Nogam Formation contains silicic volcanic clasts (Choi *et al.* 1991).

En echelon faults and folds in the southeastern part of the Bagjisan Syncline are associated with Middle–Late Triassic faulting along the Mungyeong–Jeongseon Tectonic Line (also referred to as the Geongsuweon Thrust), which is the northeastern extension of the Honam Shear Zone (Fig. 2; Reedman & Um 1975; Cluzel 1991). This tectonic line sharply cuts the western part of the Bagjisan Syncline, so that the fault motion must have occurred slightly later than the formation of the Bagjisan Syncline. The Bagjisan synclinal structure therefore formed during the Middle Triassic (Cluzel *et al.* 1991). This is confirmed by an unconformity between the Pyeongan Supergroup and the Late Triassic–Early Jurassic Daedong Supergroup. In the Bagjisan Syncline, dynamically foliated metamorphic rocks related to fault motions are only observed in peripheral southeastern areas. This fault metamorphism is of intermediate- to low-pressure type (Cluzel 1991).

160 hand samples of greenish sandstones of the Nogam Formation were collected at 23 sites in the Bagjisan Syncline for palaeomagnetic measurements (Fig. 2). One site (SK8)

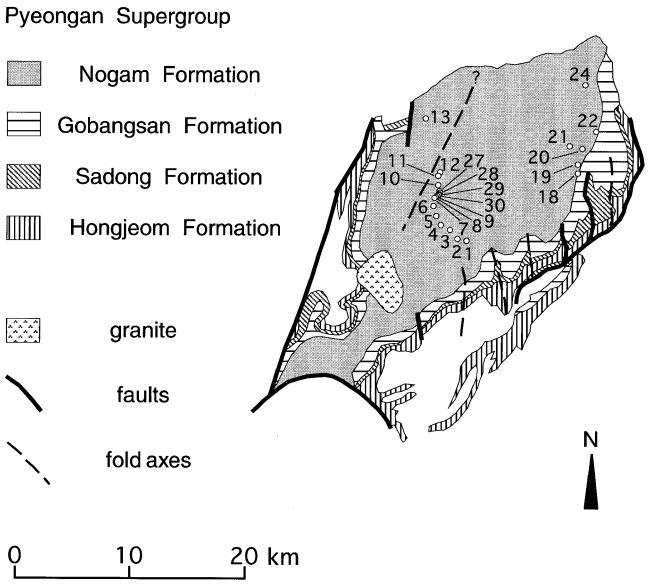


Figure 2. Geological map of the Bagjisan Syncline. Open circles denote the palaeomagnetic sampling sites in the Nogam Formation. The fold axis in the Nogam Formation is assumed from bedding attitudes in this study.

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shows a small-scale folded structure where dip varies from 24° to 77°. The other 22 sites have monoclinal structures with dips ranging from 18° to 61°. For each site, between eight and 22 hand samples were distributed over distances ranging up to 15 m and orientated with the magnetic compass. The present geomagnetic field declination was calculated using the International Geomagnetic Reference Field (IAGA Division V, Working Group 8 1995).

3 PALAEOMAGNETIC AND ROCK MAGNETIC METHODS

In the laboratory, individual samples 25 mm in diameter and 22 mm in length were prepared from each hand sample. Natural remanent magnetizations (NRMs) were measured with a 2-G cryogenic magnetometer. All samples were subjected to progressive thermal demagnetization treatment with a Natsuhara TDS-1 thermal demagnetizer in step of 50 °C up to 500 °C. Demagnetization steps of 10, 20 or 30 °C were used in the temperature range of 500–660 °C. Above 660 °C, steps of 5 °C were employed.

Results for each sample were plotted on an orthogonal vector diagram (Zijderveld 1967) to evaluate the demagnetization behaviour. Principal component analysis (Kirschvink 1980) was used to estimate the directions of the magnetic components (Table 1).

Progressive acquisition of saturation isothermal remanent magnetization (IRM) up to a maximum field of 2.7 T and thermal demagnetization of a three-component IRM (Lowrie 1990) were performed with a 2-G pulse magnetizer to identify ferromagnetic mineral components.

4 PALAEOMAGNETISM

4.1 Thermal demagnetization and rock magnetic experiments

Two magnetic components are recognized at seven sites (SK7, 8, 9, 27, 28, 29 and 30, see Figs 3a to d), while a single magnetic component is observed in eight sites (SK4, 10, 11, 12, 13, 20, 21 and 22, see Figs 3e to g and Table 1). The remaining eight sites exhibit erratic directional behaviour during thermal demagnetization, and no stable end points are identified (Fig. 3h).

4.1.1 Samples with two remanent magnetization components

A high-temperature magnetization component which decays to the origin is resolved by demagnetization steps above 500 °C. A large decay is observed in demagnetization steps above 660 °C, and the high-temperature component is unblocked between 670 and 690 °C: since 78 per cent of the samples are unblocked at 690 °C, the high-temperature component evidently resides in haematite.

The low-temperature magnetization component shows a steady decay, and 77 per cent of the samples have unblocking temperatures lower than 580 °C (Figs 3a and b). The maximum unblocking temperature of this component is 650 °C (Fig. 3d). The unblocking temperatures indicate that this component resides in either magnetite or magnetite and haematite.

IRM acquisition curves in Fig. 4(a) show low and high coercive force with a sharp initial rise up to 500 mT. Thermal demagnetization of the composite IRM shows two unblocking temperature phases, at 580 °C and 680 °C (Fig. 4b). Both magnetite and haematite are evidently magnetic carriers.

4.1.2 Samples with a single remanent magnetization component

With the exception of a component removed in the first one or two steps of treatment, these samples are dominated by a single magnetization component with unblocking temperatures of 500–680 °C. Most samples (82 per cent) show unblocking temperatures of 500–580 °C (Fig. 3e): all samples in three sites (SK4, 11 and 21) reveal this behaviour. The remaining samples show unblocking temperatures of 600–680 °C (Figs 3f and g).

Samples with unblocking temperatures of 500–580 °C show a steady decay. The remanence carrier is inferred to be magnetite from the maximum unblocking temperature of 580 °C. The IRM acquisition curve in Fig. 4(c) shows a low coercive force with a sharp initial rise up to 200 mT. Thermal demagnetization of the composite IRM shows an unblocking temperature of 580 °C (Fig. 4d), confirming that the magnetic carrier is magnetite.

Samples with unblocking temperatures of 600–680 °C have a large percentage of the initial NRM demagnetized by 580 °C. Little change is observed in the direction of the remanent magnetization after demagnetization at temperatures higher than the magnetite Curie temperature. This suggests that the remanent magnetization component resides in both magnetite and haematite. The IRM acquisition curve in Fig. 4(e) exhibits both low and high coercive force, with a sharp initial rise up to 500 mT, and thermal demagnetization of the composite IRM reveals two unblocking phases at 580 °C and 690 °C (Fig. 4f), indicating that both magnetite and haematite are magnetic carriers.

4.2 Palaeomagnetic directions

4.2.1 The high-temperature component

The site-mean directions of the high-temperature component cluster well at each site: the precision parameter of the seven site-mean directions ranges from 15.2 to 58.6. The seven directions before tilt correction with a mean of $D = 4.3°$, $I = 37.3°$ with $\alpha_{95} = 10.2°$ are distinguishable from the present direction of the Earth's magnetic field ($D = 352.3°$, $I = 53.3°$).

The mean palaeomagnetic direction of the seven sites after tilt correction ($D = 347.1°$, $I = 23.8°$ with $\alpha_{95} = 5.5°$) reveals a tight clustering (Fig. 5). The precision parameter increases from 36.1 to 120.8. The fold test is significant at the 95 per cent confidence level (McFadden 1990), indicating that there is no case for rejecting the hypothesis that the high-temperature component is of pre-folding origin. The pre-folding nature is further confirmed by a within-site fold test for the single site SK8 located within a folded structure. The precision parameter increases from 34.4 to 58.6 (Fig. 6), indicating a pre-folding origin at the 95 per cent confidence level (McFadden 1990). The high-temperature component is therefore deduced to be essentially of primary origin because folding of the Nogam Formation occurred in the Middle Triassic (Cluzel *et al.* 1991). The primary nature of the direction is further confirmed by

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Table 1. Palaeomagnetic results of the Nogam Formation.

Site	Component	N	In situ		Tilt corrected		k	α_{95} (°)	Locality		Strike/Dip	
			D (°)	I (°)	D (°)	I (°)			Lat. (°N)	Long. (°E)	(°)	(°)
SK7	HT	9	353.9	40.0	341.5	24.2	15.2	13.6	37.519	128.565	206	25
SK8	HT	22	355.5	13.2	340.3	28.4	58.6	4.1	37.519	128.566	148*	43*
SK9	HT	9	0.3	46.4	345.8	19.9	37.1	9.2	37.520	128.568	216	36
SK27	HT	15	12.5	40.9	357.3	19.4	40.6	6.1	37.526	128.568	224	33
SK28	HT	9	14.6	47.1	346.1	31.3	44.5	7.8	37.526	128.563	210	35
SK29	HT	8	11.7	30.9	348.5	26.1	22.8	11.8	37.526	128.568	186	41
SK30	HT	9	4.1	40.3	349.4	16.9	37.1	8.6	37.521	128.568	218	36
Mean	HT	7	4.3	37.3			36.1	10.2	37.500	128.500		
					347.1	23.8	120.8	5.5				
SK4	S	8	358.7	61.8	322.6	42.9	162.3	4.4	37.493	128.570	194	32
SK7	LT	17	351.3	48.3	335.8	31.0	16.0	9.2	37.519	128.565	206	25
SK8	LT	20	340.3	50.5	293.5	40.3	22.5	7.0	37.519	128.566	148*	43*
SK9	LT	10	345.0	60.3	331.0	29.3	104.7	4.7	37.520	128.568	216	36
SK10	S	8	347.2	61.3	343.1	27.5	95.0	5.7	37.526	128.567	248	34
SK11	S	5	4.0	65.4	6.3	42.4	65.4	9.5	37.535	128.567	279	23
SK12	S	7	358.9	54.3	1.2	21.4	59.8	7.9	37.537	128.569	275	33
SK13	S	5	345.1	66.5	107.4	76.4	16.7	19.3	37.581	128.554	54	33
SK20	S	8	345.6	55.3	324.0	24.3	39.0	9.0	37.559	128.703	205	39
SK21	S	8	323.4	61.3	300.2	6.5	172.4	4.2	37.556	128.716	191	61
SK22	S	8	349.2	55.8	332.2	18.8	345.2	3.0	37.569	128.732	220	42
SK27	LT	14	348.7	62.0	332.9	31.9	49.5	5.7	37.526	128.568	224	33
SK28	LT	9	337.0	61.2	315.4	29.7	54.4	7.0	37.526	128.563	210	35
SK29	LT	10	343.9	59.0	310.3	31.1	49.0	7.0	37.526	128.568	186	41
SK30	LT	10	331.7	63.6	319.8	30.1	67.4	5.9	37.521	128.568	218	36
Mean	LT + S	15	346.1	59.5			121.4	3.5	37.500	128.500		
					328.3	34.5	10.0	12.7				
SK1									37.481	128.597	139	40
SK2									37.481	128.587	136	44
SK3									37.490	128.559	152	40
SK5									37.500	128.566	223	38
SK6									37.510	128.562	245	19
SK18									37.536	128.717	197	55
SK19									37.542	128.716	204	50
SK24									37.609	128.720	222	18

Site	Component	N	In situ		Tilt corrected		k	α_{95} (°)	Locality		Strike/Dip	
			D (°)	I (°)	D (°)	I (°)			Lat. (°N)	Long. (°E)	(°)	(°)
SK8	HT	22	355.5	13.2			34.4	5.4	37.519	128.566	148*	43*
					340.3	28.4	58.6	4.1				
SK8	LT	20	340.3	50.5			40.2	5.2	37.519	128.566	148*	43*
					293.5	40.3	22.5	7.0				

HT: high-temperature component; LT: low-temperature component; S: single magnetization component; N: the number of the samples in calculation for the means; D and I: declination and inclination, respectively; k: the precision parameter; α_{95}: radius of the cone of 95 per cent confidence; Lat.: north latitude; Long.: east longitude. *Averaged bedding plane.

the obvious disagreement with the Jurassic direction from the Ryeongnam Block ($D = 31.2°$, $I = 67.4°$ with $\alpha_{95} = 8.5°$, Kim & Van der Voo 1990).

4.2.2 The low-temperature component and the single component

Palaeomagnetic directions of the low-temperature component and the single component cluster tightly before tilt correction (Fig. 7). They show normal polarity, and are close to the present direction of the Earth's magnetic field. Mean directions of the low-temperature component and the single component are indistinguishable from each other: $D = 343.0°$, $I = 58.0°$ with $\alpha_{95} = 5.2°$, and $D = 349.1°$, $I = 60.7°$ with $\alpha_{95} = 5.2°$, respectively. A mean direction of the combined 15 sites before tilt correction is $D = 346.1°$, $I = 59.5°$ with $\alpha_{95} = 3.5°$.

The directions from the 15 sites yield a negative fold test at the 95 per cent confidence level (McFadden 1990): the precision parameter decreases from 121.4 to 10.0. The within-site fold test

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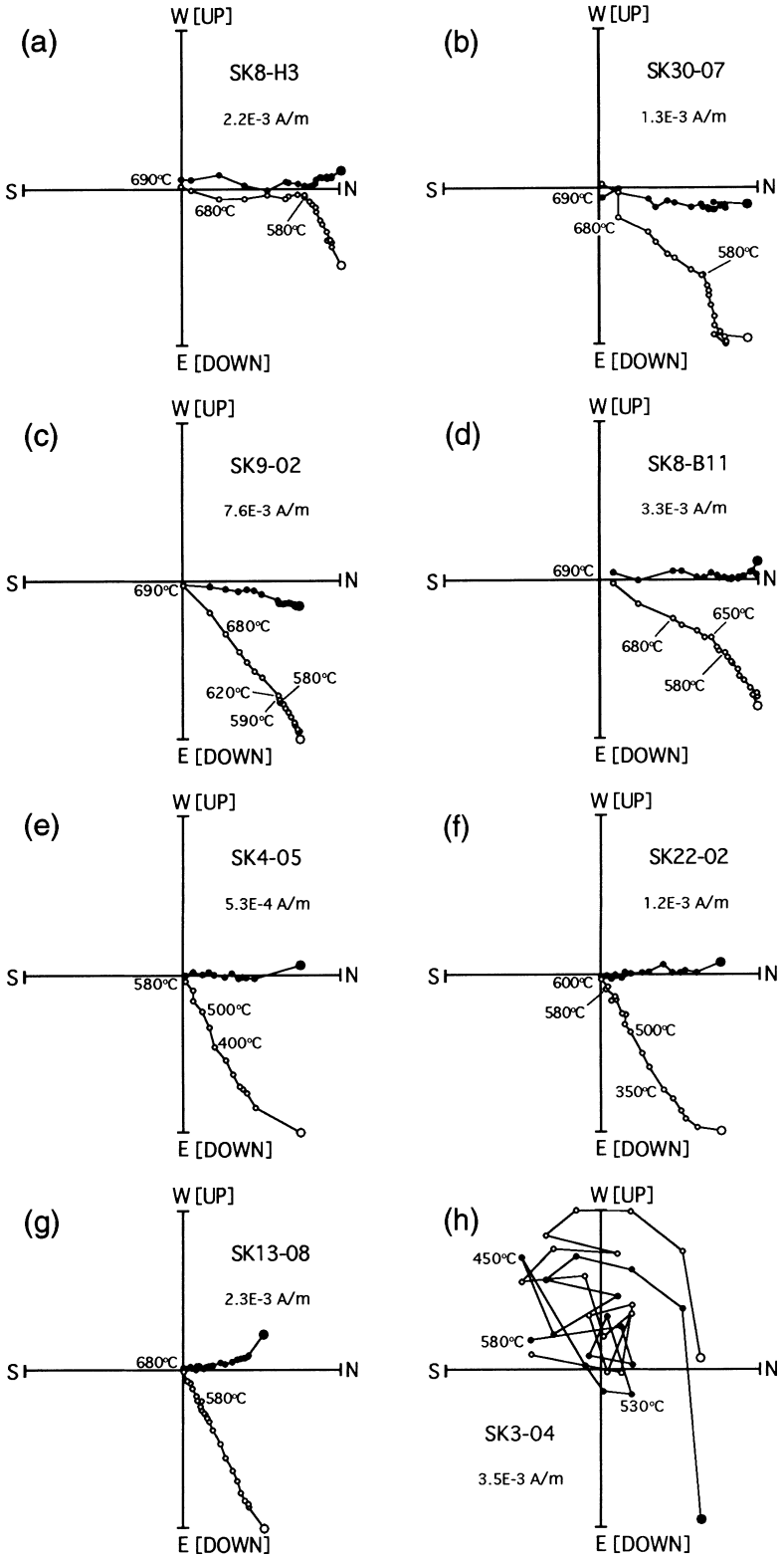


Figure 3. Orthogonal plots of thermal demagnetization of the Nogam Formation (before tilt correction). Solid and open symbols represent projection on horizontal and vertical planes, respectively.

of the low-temperature component for site SK8 also indicates a post-folding origin for this component at the 95 per cent confidence level (McFadden 1990): the precision parameter decreases from 40.2 to 22.5 (Table 1), providing no case for rejecting the hypothesis that this remanence postdates folding. These post-folding remanent magnetization were evidently acquired in Recent times, because their directions before tilt correction are close to the present field direction.

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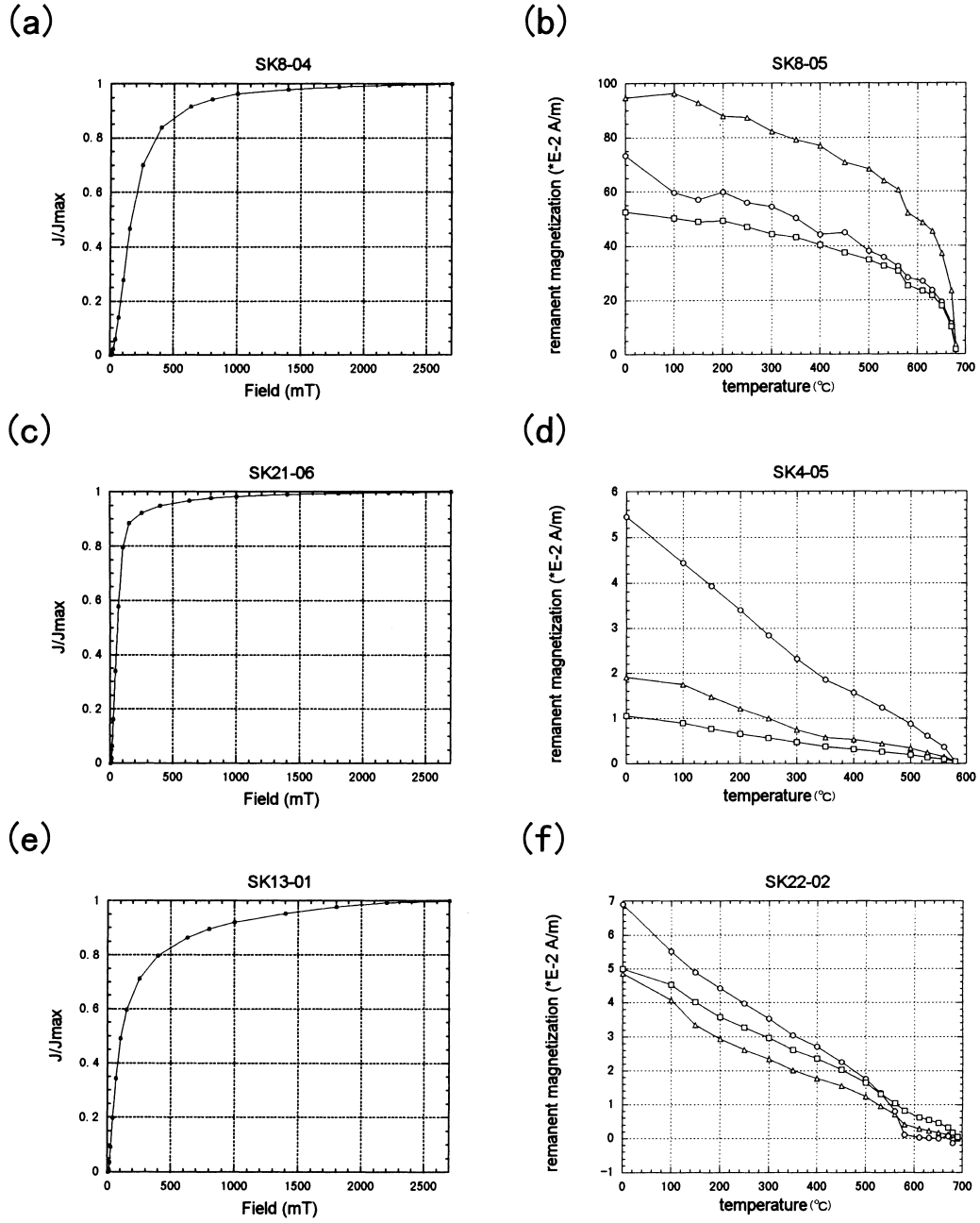


Figure 4. (a, c and e) IRM acquisition curves. (b, d and f) Thermal demagnetization of the composite IRM: circles indicate the soft component, 0.12 T field; triangles indicate the medium component, 0.4 T field; squares indicate the hard component, 2.7 T field.

5 DISCUSSION

5.1 Remanent magnetization of the Nogam Formation

The Nogam Formation in the Bagjisan Syncline yields a primary component following removal of a low-temperature component. The component resides in haematite, indicated by unblocking temperatures of 670–690 °C. This component in the Nogam Formation is obtained from a stratigraphic thickness of about 100 m, although the direction contains a single normal polarity. Ogg & Steiner (1991) revealed normal polarity zones covering a stratigraphic sequence of about 100 m in Lower Triassic sediments, and found that normal polarity is predominant in Smithian times (late Early Triassic). The remanence of the Nogam Formation can be interpreted as a record of this period. The mean direction of the Nogam Formation is therefore regarded as an Early Triassic characteristic direction for the Ryeongnam Block.

Sampling sites with the primary magnetization are located on the central part of the Bagjisan Syncline (Fig. 2). The central part of the Bagjisan Syncline escaped remagnetization, whereas other parts of the syncline were completely remagnetized. Since the peripheral area of the syncline suffered Mesozoic fault deformation and emplacement of granitic rocks (Reedman

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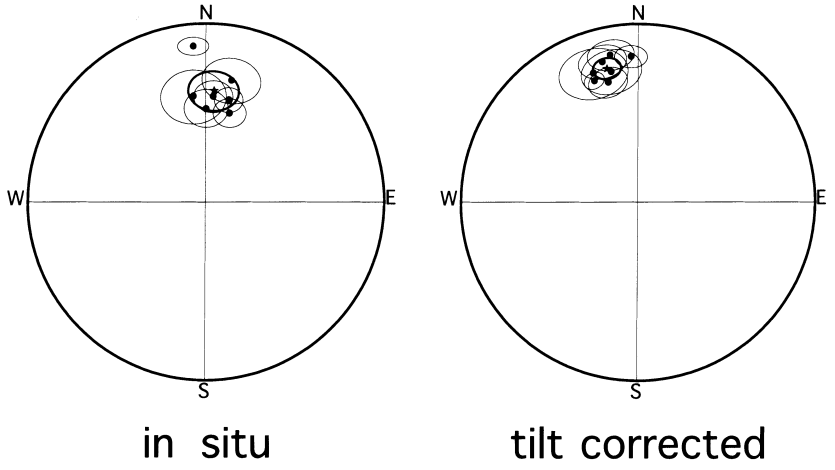


Figure 5. Equal-area projections of the directions of the high-temperature components (with associated 95 per cent confidence limits) from seven sites of the Nogam Formation. The star and bold circle represent the mean direction and associated 95 per cent confidence limit, respectively. The fold test is positive at the 95 per cent confidence level (McFadden 1990).

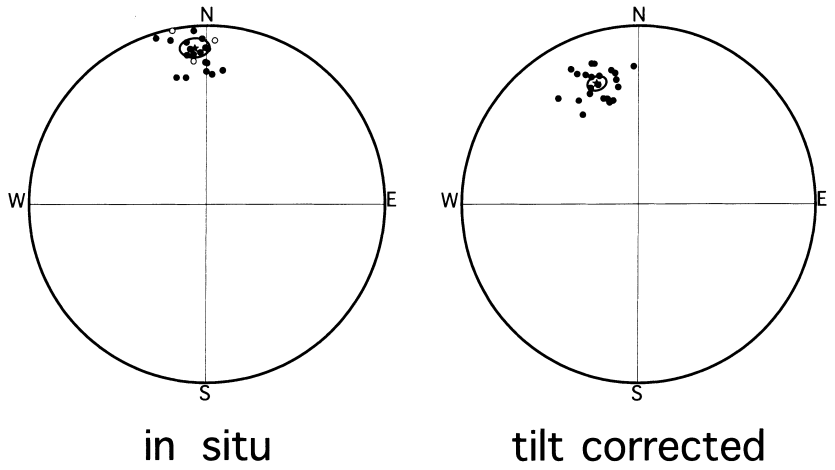


Figure 6. Equal-area projections of the directions of the high-temperature components for site SK8 from 22 samples. The folded structure of the site enables a within-site fold test to be performed. The star and circle are the mean direction and associated 95 per cent confidence limit, respectively. The fold test is positive at the 95 per cent confidence level (McFadden 1990).

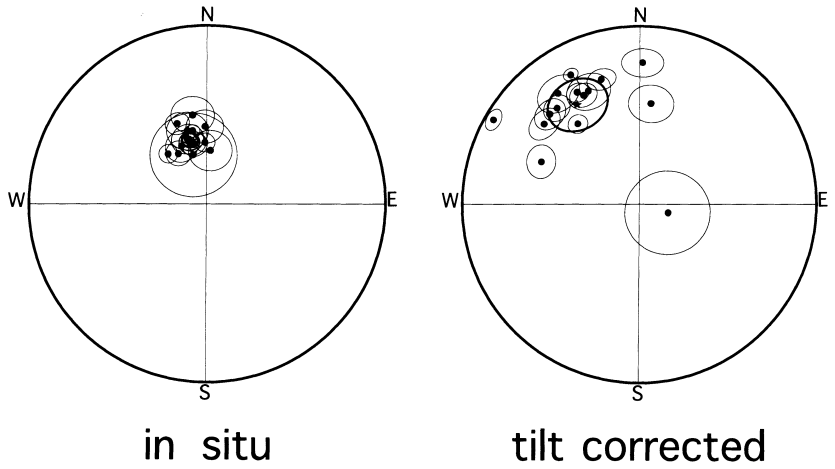


Figure 7. Equal-area projections of the palaeomagnetic directions of the low-temperature components and the single magnetization components (with associated 95 per cent confidence limits) from 15 sites in the Nogam Formation. The star and bold circle represent the mean direction and associated 95 per cent confidence limit, respectively. The fold test is negative at the 95 per cent confidence level (McFadden 1990).

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& Um 1975; Lee 1987), deformation leading to fracturing and higher permeability may have facilitated remagnetization by recent weathering.

5.2 Coincidence of palaeomagnetic poles between the Ryeongnam Block and the North China Block

The Early Triassic palaeomagnetic pole for the Ryeongnam Block (Table 2) is derived from the seven primary site-mean directions from the Nogam Formation (62.5°N, 336.8°E with $A_{95} = 4.7°$) and is compared with the Early Triassic pole for the North China Block at 57.7°N, 352.8°E with $A_{95} = 6.3°$ (Fig. 8; Yang *et al.* 1991). The pole for the Ryeongnam Block plots to the west of the contemporaneous pole from the North China Block. Although the difference between the two poles ($9.3° \pm 7.9°$) is significant, there is good agreement. A palaeolatitude contrast would be recognized by a difference between the observed palaeolatitude and the expected one derived from the pole for the North China Block at the reference point within the Ryeongnam Block (37.5°N, 128.5°E): no significant palaeolatitude difference ($0.2° \pm 7.9°$) is observed.

In contrast to the case for the North China Block, the Early Triassic pole for the South China Block (50.5°N, 226.1°E with $A_{95} = 6.5°$, Huang & Opdyke 1996) is situated far from the pole for the Ryeongnam Block (Fig. 8). The angular distance between the two poles is $54.5° \pm 8.0°$. A significant palaeolatitude difference of $11.3° \pm 8.0°$ is also present between the observed palaeolatitude and the expected one derived from the pole for the South China Block at the reference point of the Ryeongnam Block.

Table 2. Early Triassic palaeomagnetic poles for the Ryeongnam, North China and South China Blocks.

Locality	Age	N (*n*)	Pole		A_{95} (°)	References
			Lat. (°N)	Long. (°E)		
RB (BJS)	Tr1	(7)	62.5	336.8	4.7	This study
RB (BGS)	Tr1	(4)	63.2	306.1	12.6	Doh & Piper (1994)
RB (BGS)	Tr1	3	26.4	213.9	15.6	Kim & Van der Voo (1990), Kim *et al.* (1992)
RB (BGS)	Tr1	(3)	9.4	203.1	19.7	Lee *et al.* (1997)
NCB	Tr1	3	57.7	352.8	6.3	Yang *et al.* (1991)
SCB	Tr1	4	50.5	226.1	6.5	Huang & Opdyke (1996)

RB: Ryeongnam Block; NCB: North China Block; SCB: South China Block; BJS: Bagjisan Syncline; BGS: Baegunsan Syncline; Tr1: Early Triassic; N (*n*): the number of the studies or areas (sites); A_{95}: radius of the circle of 95 per cent confidence; Lat.: north latitude; Long.: east longitude.

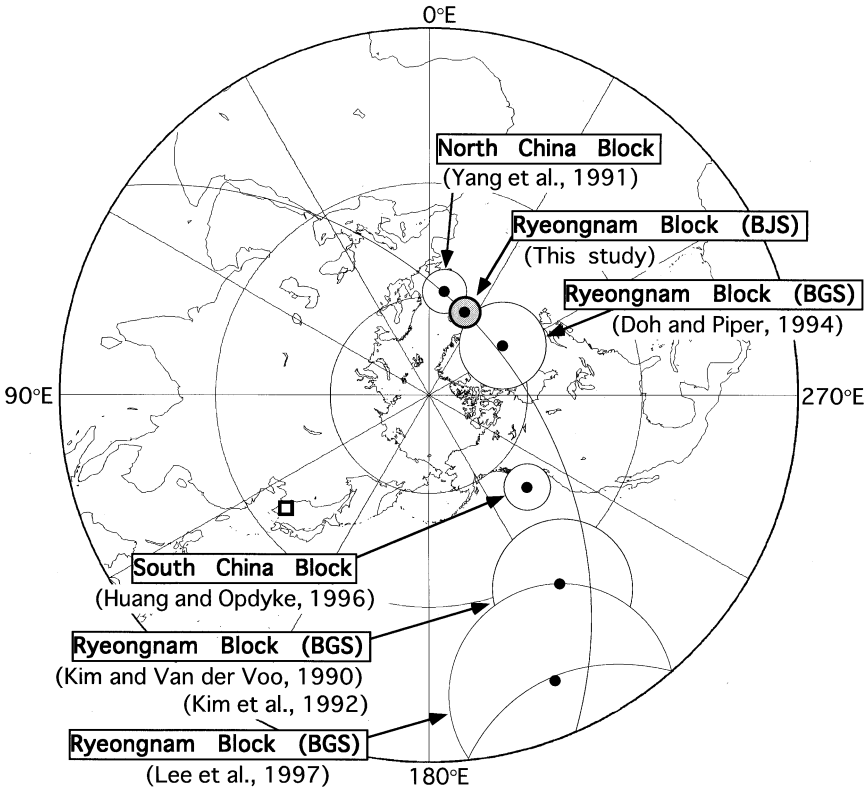


Figure 8. Early Triassic palaeomagnetic poles (with associated 95 per cent confidence limits) for the Ryeongnam Block, North China Block and the South China Block. BJS and BGS are the Bagjisan and Baegunsan synclines, respectively. The open square denotes the reference point of the Ryeongnam Block. Small circle is centred on the reference point. Data are from Table 2.

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The Ryeongnam Block seems to have rotated slightly clockwise with respect to the North China Block since the Early Triassic, although the amount of the rotation is not quite significant ($9.6^\circ \pm 10.0^\circ$) at the 95 per cent confidence level at the reference point within the Ryeongnam Block. A small difference in pole positions between the Ryeongnam Block and the North China Block can be explained by post-Cretaceous clockwise rotation (10°–29°) of the area east of the Tan-Lu Fault with respect to the main part of the North China Block, as suggested by Ma *et al.* (1993). Restoration of this clockwise rotation gives a better coincidence of the palaeomagnetic poles between the Ryeongnam Block and the North China Block. It is therefore concluded that the Ryeongnam Block has behaved as an integral part of the North China Block since the Early Triassic.

5.3 Differential clockwise rotation in the Ryeongnam Block

The previously reported Early Triassic palaeomagnetic directions of the Ryeongnam Block have variable declinations, whereas inclinations are fairly consistent values of about 25°. Doh & Piper (1994) reported a declination of 1.1°, while Lee *et al.* (1996) presented a declination of 91.9°. A declination of 70.9° is recalculated from palaeomagnetic results of Kim & Van der Voo (1990) and Kim *et al.* (1992). These authors demonstrated the primary nature of the palaeomagnetic directions by positive fold tests and divergence of the palaeomagnetic directions between Early Triassic and younger ages. Most of these data were collected from the Baegunsan Syncline located about 30 km away from the Bagjisan Syncline of this study (Fig. 1b).

The palaeomagnetic poles for the Ryeongnam Block from previous studies have fairly large confidence limits (Fig. 8, Table 2). These poles, however, coincide neither with the pole for the North China Block nor with that for the South China Block. Furthermore, a large uncertainty of the poles invalidates the comparison of palaeolatitudes among the poles from the Ryeongnam and Chinese blocks, because the palaeolatitude difference between the North and South China Blocks differs only by $11.5^\circ \pm 9.1^\circ$ at the reference point within the Ryeongnam Block. The Ryeongnam Block pole in the present study is therefore the first reliable pole that enables the comparison of pole positions and palaeolatitude difference.

In spite of the large uncertainty of the poles, it is noticeable that the poles from the previous studies reveal clockwise deflection relative to the pole of the present study. The palaeomagnetic poles for the Ryeongnam Block plot along a small circle centred on the reference point within the Ryeongnam Block (Fig. 8, Table 2) and the small circle passes through the Early Triassic pole for the North China Block. The pole of the present study is not significantly rotated clockwise with respect to the North China Block ($9.6^\circ \pm 10.0^\circ$), whereas the poles of Doh & Piper (1994), Kim & Van der Voo (1990) and Kim *et al.* (1992), and Lee *et al.* (1997) show clockwise rotations of 23.5°, 93.3°, and 114.2° with respect to the North China Block, respectively.

The clockwise-deflected directions observed in the Ryeongnam Block can be explained either by tectonic rotation or by partial remagnetization. Because a primary origin was demonstrated by positive fold tests in the previous studies, it would appear that tectonic rotation is responsible for the clockwise rotations. According to structural analysis around the Baegunsan Syncline (Cluzel 1991), many NNE-trending vertical faults cut the syncline, and faulting along faults with this trend has been associated with the Triassic faulting along the Honam Shear Zone (Cluzel 1991). The variable declinations derived from the Baegunsan Syncline are therefore ascribed to differential local rotations due to the dextral fault motion along the Honam Shear Zone. The Triassic faulting along the Honam Shear Zone is considered to be the result of the collision between the North and South China blocks (Yin & Nie 1993; Li 1994), because the collisional event is the only predominant tectonic event for the Ryeongnam Block in this period.

The differential rotation mode in the Baegunsan Syncline is further confirmed by the original results of Doh & Piper (1994). They rejected three out of seven Early Triassic sites in the Baegunsan Syncline because of dispersion in palaeomagnetic direction. These three sites show variable clockwise-rotated declinations with consistent inclinations relative to the remaining four sites. The original results of Doh & Piper (1994) revealed differential rotation in the Baegunsan Syncline.

In the Bagjisan Syncline, the distribution of the faults is restricted to the southeastern periphery (Fig. 2; Cluzel 1991). Sampling sites with primary directions are free from the fault deformation.

5.4 Implications for the collision of the North and South China blocks in the Korean Peninsula

Palaeomagnetic poles for the North China Block and the South China Block have been established from the Permian onwards (Enkin *et al.* 1992). The palaeomagnetic poles for the North China Block in Permian–Triassic times lie around 0°E longitude, 50°–60°N latitude, while the poles for the South China Block in Permian–Triassic times fall around 200°–250°E longitude, 40°–50°N latitude. This shows that the North China Block and the South China Block were separate from each other in Permian–Triassic times. The separation of the poles between the two blocks disappears in the Late Jurassic (Gilder & Courtillot 1997), indicating that the North and South China blocks were completely united by this time.

The North and South China blocks have different Palaeozoic bioprovinces. The bioprovinces of the two blocks are bounded by the Dabie and Shandong belts in China (Fig. 1a; Kobayashi 1966; Kobayashi 1967). This suggests that the North and the South China blocks were located in geographically different positions during Palaeozoic times. The welding of the two blocks occurred in the Dabie and Shandong belts (Okay & Sengör 1992).

The present study indicates that the southeastern tip of the Korean Peninsula (the Ryeongnam Block) belongs to the North China Block. However, the central part of the peninsula has been correlated with the South China Block from palaeontological data (Kobayashi 1967). This correlation requires that welding between the North and South China blocks occurred in the central part of the present Korean Peninsula. This interpretation implies that the southeastern part of the Korean Peninsula is now geographically isolated from the main part of the North China Block, and therefore that the South China Block was obducted onto the North China Block at the site of the present Korean Peninsula during the welding event. Geological observations have revealed that the two Palaeozoic bioprovinces of the North and South China blocks have a

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thrust contact along the Geongsuweon Thrust within the Korean Peninsula (Reedman & Um 1975), suggesting obduction in the Korean Peninsula.

This interpretation is supported by a crustal-detachment structure observed around the Korean Peninsula. Li (1994) suggested that the upper crust of the South China Block to the east of the Tan-Lu Fault was detached from the lower crust, and thrust over the North China Block for more than 400 km. The northern margin of the North China Block shows sinistral displacement of only 150 km along the Tan-Lu Fault (Xu *et al.* 1987). The discrepancy between these displacements suggests that the Gyeonggi Block was conveyed onto the Korean Peninsula on a low-angle detachment. The likely depth of this possible detachment was estimated to be $\sim$20 km by Li (1994).

The data sets of palaeomagnetic poles for the North and South China blocks indicate that the North China Block has rotated counterclockwise more than 60° relative to the South China Block since Triassic times, and that the two blocks sutured following a scissor-like collision process (Zhao & Coe 1987). The collision of the two blocks occurred initially in their easternmost parts and progressed westwards (Fig. 9). With this model, the Korean Peninsula is the initial collision area between the North and South China blocks. Since the easternmost part of the North and South China blocks is the Korean Peninsula, obduction of the South China Block onto the North China Block in the Korean Peninsula is the initial collisional event. This westward progression is also observed in the sedimentary facies transition from marine carbonates to terrestrial clastics, which records closure of an ocean between the North and South China blocks (Yin & Nie 1993). The collisional areas between the North and South China blocks are the sites of Permian–Triassic fold belts in Korea and China. These fold belts extend from the Honam Shear Zone to the Imjingang Belt, the Shandong Belt and the Dabie Belt from east to west (Fig. 1a). The facies transition in Korea commenced in the Early Permian, while that in China occurred between the Late Permian and Middle Triassic (Yin & Nie 1993).

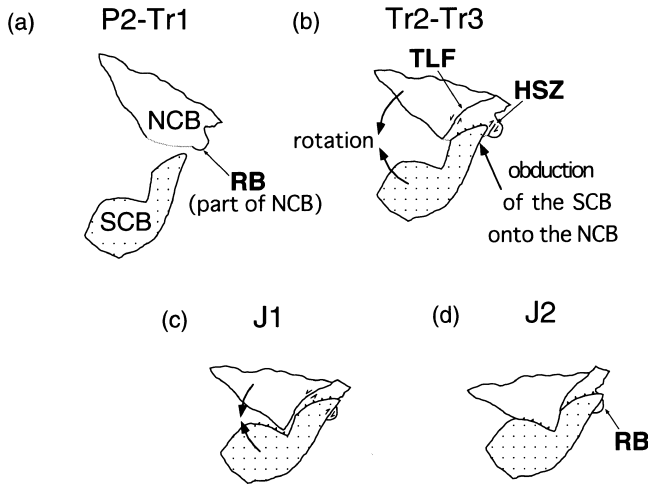


Figure 9. Tectonic model for the process of the collision between the North and South China blocks (modified from Yin & Nie 1993). The collision occurred initially in the Korean Peninsula and progressed westwards through a scissor-like process (Zhao & Coe 1987). TLF: Tan-Lu Fault; HSZ: Honam Shear Zone; RB: Ryeongnam Block.

The collision has apparently isolated the Ryeongnam Block from the main part of the North China Block without any spatial displacement of the Ryeongnam Block (Fig. 9). However, the collision of the South China Block into the North China Block gave rise to a northward displacement of the area between the Tan-Lu Fault and the Honam Shear Zone in the North China Block, because no palaeolatitude difference ($0.2° \pm 7.9°$) is observed between the Ryeongnam Block and the interior part of the North China Block and because the northern margin of the North China Block shows sinistral displacement of about 150 km along the Tan-Lu Fault (Xu *et al.* 1987).

6 CONCLUSIONS

(1) The Early Triassic Nogam Formation in the Bagjisan Syncline has yielded a reliable remanent magnetization component with unblocking temperatures of 670–690 °C. This component resides in haematite.

(2) The high-temperature component from seven sites of the Nogam Formation is probably of primary origin, as indicated by a positive fold test, indicating that remanence predates Middle Triassic folding. The mean direction of the primary component ($D = 347.1°$, $I = 23.8°$ with $\alpha_{95} = 5.5°$) is regarded as the characteristic direction for the Ryeongnam Block in the Early Triassic.

(3) The palaeomagnetic pole derived from the primary component (62.5°N, 336.8°E with $A_{95} = 4.7°$) is comparable to the Early Triassic pole for the North China Block. The Ryeongnam Block has therefore been an integral part of the North China Block since early Mesozoic times.

(4) The Ryeongnam Block, the southern sector of the Korean Peninsula, is an eastern part of the North China Block. The Ryeongnam Block has been geographically isolated from the main part of the North China Block by obduction of the Gyeonggi Block onto the central part of the Korean Peninsula.

(5) The collision between the North and South China blocks occurred initially at the Korean Peninsula and progressed westwards.

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